

Linear algebra: Exercise session 4

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1. Assume that $A \in \mathbb{R}^{m \times n}$. Show that the following four fundamental sets are subspaces:

$$C(A) = \{Ax : x \in \mathbb{R}^n\}, \quad C(A^T) = \{A^T x : x \in \mathbb{R}^m\}, \quad N(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \quad N(A^T) = \{x \in \mathbb{R}^m : A^T x = 0\}.$$

Answer.

Column space of A:

- First, $0 = A0 \in C(A)$.
- If $Ax_1, Ax_2 \in C(A)$, where $x_1, x_2 \in \mathbb{R}^n$, then

$$Ax_1 + Ax_2 = A(x_1 + x_2) \in C(A).$$

- If $Ax \in C(A)$, where $x \in \mathbb{R}^n$, and $\lambda \in \mathbb{R}$, then

$$\lambda(Ax_1) = A(\lambda x_1) \in C(A).$$

Hence, $C(A)$ is a subspace of $\mathbb{R}^m$.

Row space of A:

- First, $0 = A^T 0 \in C(A^T)$.
- If $A^T x_1, A^T x_2 \in C(A^T)$, where $x_1, x_2 \in \mathbb{R}^m$, then

$$A^T x_1 + A^T x_2 = A^T(x_1 + x_2) \in C(A^T).$$

- If $A^T x \in C(A^T)$, where $x \in \mathbb{R}^m$, and $\lambda \in \mathbb{R}$, then

$$\lambda(A^T x_1) = A^T(\lambda x_1) \in C(A^T).$$

Hence, $C(A^T)$ is a subspace of $\mathbb{R}^n$.

Nullspace of A:

- First, $A0 = 0$, hence $0 \in N(A)$.
- If $x_1, x_2 \in N(A)$, then

$$A(x_1 + x_2) = Ax_1 + Ax_2 = 0 + 0 = 0,$$

meaning that $x_1 + x_2 \in N(A)$.

- If $x \in N(A)$ and $\lambda \in \mathbb{R}$, then

$$A(\lambda x) = \lambda(Ax) = 0,$$

meaning that $\lambda x \in N(A)$.

Therefore, $N(A)$ is a subspace of $\mathbb{R}^n$.

Left nullspace of A^T:

- First, $A^T 0 = 0$, hence $0 \in N(A^T)$.
- If $x_1, x_2 \in N(A^T)$, then

$$A^T(x_1 + x_2) = A^T x_1 + A^T x_2 = 0 + 0 = 0,$$

meaning that $x_1 + x_2 \in N(A^T)$.

- If $x \in N(A^T)$ and $\lambda \in \mathbb{R}$, then

$$A^T(\lambda x) = \lambda(A^T x) = 0,$$

meaning that $\lambda x \in N(A^T)$.

Therefore, $N(A^T)$ is a subspace of $\mathbb{R}^m$.

2. Find bases for four fundamental spaces of the following matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 3 & 8 & 6 & 16 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 2 \\ 1 & -2 \\ 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 & 1 & 5 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & -1 & 1 \end{bmatrix}.$$

Answer.

Matrix A:

$$[A|0] = \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 8 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 - 3r_1} \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 2 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow \frac{1}{2}r_2} \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right].$$

$$C(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \end{bmatrix} \right\} = \mathbb{R}^2, \quad C(A^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2.$$

$$N(A) = \{x \in \mathbb{R}^2 : Ax = 0\} = \{0\}, \quad N(A^T) = \{x \in \mathbb{R}^2 : A^T x = 0\} = \{0\}.$$

$$\dim(C(A)) = \dim(C(A^T)) = \text{rank}(A) = 2, \quad \dim(N(A)) = 0, \quad \dim(N(A^T)) = 0.$$

Matrix B:

$$[B|0] = \left[\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 0 \\ 3 & 8 & 6 & 16 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 - 3r_1} \left[\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 0 \\ 0 & 2 & 0 & 4 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow \frac{1}{2}r_2} \left[\begin{array}{cccc|c} 1 & 2 & 2 & 4 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{array} \right] \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{array} \right].$$

$$C(B) = \text{span} \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 8 \end{bmatrix} \right\}, \quad C(B^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 2 \end{bmatrix} \right\}.$$

$$N(B) = \{x \in \mathbb{R}^4 : Bx = 0\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} : \begin{array}{l} x_1 = -2x_3, \\ x_2 = -2x_4 \end{array} \right\} = \text{span} \left\{ \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right\}, \quad N(B^T) = \{0\}.$$

$$\dim(C(B)) = \dim(C(B^T)) = \text{rank}(B) = 2, \quad \dim(N(B)) = 2, \quad \dim(N(B^T)) = 0.$$

Matrix C:

$$[C|0] = \left[\begin{array}{cc|c} -1 & 2 & 0 \\ 1 & -2 & 0 \\ 1 & 0 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow r_2 + r_1} \left[\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{array} \right] \xrightarrow{r_3 \rightarrow r_3 + r_1} \left[\begin{array}{cc|c} -1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 2 & 0 \end{array} \right] \\ \xrightarrow[r_1 \rightarrow -r_1]{r_3 \rightarrow \frac{1}{2}r_3} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{r_1 \rightarrow r_1 + 2r_3} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow r_3} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

$$C(C) = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix} \right\}, \quad C(C^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

$$N(C) = \{x \in \mathbb{R}^2 : Cx = 0\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 = 0, x_2 = 0 \right\} = \{0\}.$$

$$[C^T|0] = \left[\begin{array}{ccc|c} -1 & 1 & 1 & 0 \\ 2 & -2 & 0 & 0 \end{array} \right] \xrightarrow[r_1 \rightarrow -r_1]{r_2 \rightarrow 2r_1 + r_2} \left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \xrightarrow{r_2 \rightarrow \frac{1}{2}r_2} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$N(C^T) = \{x \in \mathbb{R}^3 : C^T x = 0\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : \begin{array}{l} x_1 = x_2, \\ x_3 = 0 \end{array} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

$$\dim(C(C)) = \dim(C(C^T)) = \text{rank}(C) = 2, \quad \dim(N(C)) = 0, \quad \dim(N(C^T)) = 1.$$

Matrix D:

$$[D|0] = \left[\begin{array}{cccc|c} 1 & 2 & 1 & 5 & 0 \\ 2 & 4 & -3 & 0 & 0 \\ 1 & 2 & -1 & 1 & 0 \end{array} \right] \xrightarrow[r_2 \rightarrow r_2 - 2r_1]{r_3 \rightarrow r_3 - r_1} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 5 & 0 \\ 0 & 0 & -5 & -10 & 0 \\ 0 & 0 & -2 & -4 & 0 \end{array} \right]$$

$$\xrightarrow{r_2 \rightarrow -\frac{1}{5}r_2} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 5 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & -2 & -4 & 0 \end{array} \right] \xrightarrow[r_3 \rightarrow r_3 - 2r_2]{r_1 \rightarrow r_1 - r_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

$$C(D) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ -1 \end{bmatrix} \right\}, \quad C(D^T) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}.$$

$$N(D) = \{x \in \mathbb{R}^4 : Dx = 0\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} : \begin{array}{l} x_1 + 2x_2 + 3x_4 = 0, \\ x_3 + 2x_4 = 0 \end{array} \right\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} : \begin{array}{l} x_1 = -2x_2 - 3x_4, \\ x_3 = -2x_4 \end{array} \right\} = \text{span} \left\{ \begin{bmatrix} -3 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

$$[D^T|0] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \\ 1 & -3 & -1 & 0 \\ 5 & 0 & 1 & 0 \end{array} \right] \xrightarrow[r_2 \rightarrow r_2 - 2r_1]{r_3 \rightarrow r_3 - r_1, r_4 \rightarrow r_4 - 5r_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -5 & -2 & 0 \\ 0 & -10 & -4 & 0 \end{array} \right] \xrightarrow{r_2 \leftrightarrow r_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -5 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -10 & -4 & 0 \end{array} \right]$$

$$\xrightarrow{r_2 \rightarrow -\frac{1}{5}r_2, r_4 \rightarrow r_4 - 2r_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{5} & 0 \\ 0 & 1 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$N(D^T) = \{x \in \mathbb{R}^3 : D^T x = 0\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : \begin{array}{l} x_1 + \frac{1}{5}x_3 = 0, \\ x_2 + \frac{2}{5}x_3 = 0 \end{array} \right\} = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : \begin{array}{l} x_1 = -\frac{1}{5}x_3, \\ x_2 = -\frac{2}{5}x_3 \end{array} \right\} = \text{span} \left\{ \begin{bmatrix} -\frac{1}{5} \\ -\frac{2}{5} \\ 1 \end{bmatrix} \right\}.$$

$$\dim(C(D)) = \dim(C(D^T)) = \text{rank}(D) = 2, \quad \dim(N(D)) = 2, \quad \dim(N(D^T)) = 1.$$

3. Let $A \in \mathbb{R}^{m \times n}$ with $m \leq n$ and system $Ax = b$ is solvable for every $b \in \mathbb{R}^n$. Describe the column and left nullspace and calculate the dimension of four fundamental spaces.

Answer.

For all $b \in \mathbb{R}^n$ there exists a vector $x \in \mathbb{R}^m$ such that $b = Ax \in C(A)$. Therefore, the column space of A must be the entire $\mathbb{R}^m$ which meaning $C(A) = \mathbb{R}^m$ and $N(A^T) = \{0\}$. Thus,

$$\dim(C(A)) = \dim(C(A^T)) = \text{rank}(A) = m, \quad \dim(N(A)) = n - m, \quad \dim(N(A^T)) = 0.$$

4. Let $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$ and the system $Ax = b$ has a unique solution. Describe the column, row and (left) null spaces and calculate the dimension of four fundamental spaces.

Answer.

This means that the reduced row echelon matrix of A is identity matrix. Thus, $C(A) = C(A^T) = \mathbb{R}^n$ and $N(A) = N(A^T) = \{0\}$, and

$$\dim(C(A)) = \dim(C(A^T)) = \text{rank}(A) = n, \quad \dim(N(A)) = \dim(N(A^T)) = 0.$$

5. Construct a 3×3 matrix A whose column space contains $(1, 1, 0)^T$ and $(1, 0, 1)^T$ but not $(1, 1, 1)^T$.

Answer.

$$M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

6. Construct a 3×3 matrix A whose column space is only a line.

Answer.

$$N = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

7. Let A is a 7×9 matrix with $\text{rank}(A) = 5$. What are the dimensions of the four subspaces? What is the sum of all four dimensions?

Answer.

$$\dim(C(A)) = \dim(C(A^T)) = \text{rank}(A) = 5.$$

$$\dim(N(A)) = n - \text{rank}(A) = 9 - 5 = 4.$$

$$\dim(N(A^T)) = m - \text{rank}(A) = 7 - 5 = 2.$$

$$\dim(C(A)) + \dim(C(A^T)) + \dim(N(A)) + \dim(N(A^T)) = 5 + 5 + 4 + 2 = 16.$$

8. Let A is a 3×4 matrix with $\text{rank}(A) = 3$. What are its column space and left nullspace and the dimension of four fundamental subspaces?

Answer.

$$\dim(C(A)) = \dim(C(A^T)) = \text{rank}(A) = 3.$$

$$\dim(N(A)) = n - \text{rank}(A) = 4 - 3 = 1.$$

$$\dim(N(A^T)) = m - \text{rank}(A) = 3 - 3 = 0.$$

$$\text{Thus, } C(A) = \mathbb{R}^3 \text{ and } N(A^T) = \{0\}.$$

9. Does the set of all 3×3 matrices of rank 3 forms a linear space? How about the set of all 3×3 matrices of rank less than or equal to 2?

Answer.

The set of all 3×3 matrices of rank 3 does not contain zero matrix. Thus, this set is not a subspace. Furthermore, the set of all 3×3 matrices of rank less than or equal to 2 is not a subspace as well. Because elementary matrices E_{11} , E_{22} , and E_{33} are 3×3 matrices of rank 1 and belong in this set, but the addition of them is identity matrix with rank 3 and does not belong in this set. So, this set is not closed with respect to addition operator.
